

GENETICS & MOLECULAR INHERITANCE

A Concise Guide to DNA Structure, Gene Expression,
and the Rules of Heredity

An Illustrated Primer

1. The Structure of DNA

Deoxyribonucleic acid (DNA) is the molecule that stores hereditary information in nearly all living organisms. It is built from repeating units called nucleotides, each composed of a phosphate group, a deoxyribose sugar, and one of four nitrogenous bases: adenine (A), thymine (T), guanine (G), and cytosine (C).

The Double Helix

James Watson and Francis Crick, building on X-ray diffraction data from Rosalind Franklin and Maurice Wilkins, proposed in 1953 that DNA forms a double helix — two strands wound around a common axis, held together by hydrogen bonds between complementary base pairs. Adenine always pairs with thymine (two hydrogen bonds), and guanine always pairs with cytosine (three hydrogen bonds). This complementarity is the chemical basis for accurate copying of genetic information.

Base Pair	Bonds	Strand Orientation
A – T	2 hydrogen bonds	Antiparallel (5'→3' / 3'←5')
G – C	3 hydrogen bonds	Antiparallel (5'→3' / 3'←5')

DNA Packaging

In eukaryotic cells, DNA does not float freely — it wraps around histone proteins to form nucleosomes, which coil further into chromatin fibers and ultimately condense into chromosomes during cell division. This packaging allows roughly two meters of DNA to fit inside a nucleus just a few micrometers wide, while still permitting selective access to genes when they are needed.

Replication

Before a cell divides, its DNA must be copied. The enzyme helicase unwinds the double helix, and DNA polymerase synthesizes new complementary strands using each original strand as a template. Because each daughter molecule contains one old strand and one new strand, this process is called semi-conservative replication. Proofreading by DNA polymerase keeps the error rate extremely low, though occasional mismatches become the raw material for mutation and evolution.

2. From Gene to Protein: Expression

A gene is a segment of DNA that carries the instructions for building a functional product — usually a protein. Converting that instruction into a working molecule happens in two major stages: transcription and translation, together known as the central dogma of molecular biology (DNA → RNA → protein).

Transcription

Inside the nucleus, RNA polymerase binds to a promoter region and unwinds a short stretch of DNA. It reads one strand (the template strand) and synthesizes a complementary strand of messenger RNA (mRNA), substituting uracil (U) for thymine. In eukaryotes, this initial transcript is processed: non-coding introns are removed by the spliceosome, and a protective 5' cap and 3' poly-A tail are added before the mature mRNA leaves the nucleus.

Translation

In the cytoplasm, ribosomes read the mRNA sequence three bases at a time — each triplet, or codon, specifies one amino acid according to the genetic code. Transfer RNA (tRNA) molecules ferry the matching amino acids to the ribosome, which links them together into a growing polypeptide chain. A start codon (AUG) initiates translation, and one of three stop codons terminates it. The resulting chain folds into a functional protein, often with help from chaperone proteins.

- The genetic code is **degenerate**: most amino acids are specified by more than one codon.
- The code is **nearly universal** across life, reflecting a shared evolutionary origin.
- **Gene regulation** (via promoters, enhancers, transcription factors, and epigenetic marks such as DNA methylation) determines when and how much a gene is expressed.
- Not all genes are expressed at once — a skin cell and a neuron carry the same DNA but express very different genes.

Mutation

Changes in the DNA sequence — substitutions, insertions, or deletions — can alter the resulting protein. A single base substitution may be silent (no change to the amino acid), missense (a different amino acid), or nonsense (a premature stop codon). Insertions or deletions that are not multiples of three cause a frameshift, scrambling every codon downstream. While many mutations are neutral or harmful, some introduce the genetic variation that natural selection acts upon.

3. Patterns of Molecular Inheritance

Inheritance describes how genetic information passes from parents to offspring. Gregor Mendel's 19th-century pea plant experiments established the basic rules, which modern molecular genetics later explained in terms of chromosomes, alleles, and DNA.

Mendelian Principles

Each individual carries two copies (alleles) of a gene, one inherited from each parent. The **Law of Segregation** states that these two alleles separate during gamete formation, so each sperm or egg carries only one. The **Law of Independent Assortment** states that alleles of different genes, located on different chromosomes, are distributed to gametes independently of one another.

Pattern	Description	Example
Dominant/Recessive	One allele masks the other in heterozygotes	Brown vs. blue eye color genes
Incomplete dominance	Heterozygote shows a blended phenotype	Pink flowers from red x white
Codominance	Both alleles are fully expressed	AB blood type
X-linked	Genes carried on the X chromosome	Red-green color blindness
Polygenic	Multiple genes jointly influence a trait	Human height, skin tone

Beyond Mendel: Molecular Complications

Not all inheritance follows simple Mendelian ratios. **Linked genes** that sit close together on the same chromosome tend to be inherited as a unit unless separated by crossing over during meiosis. **Epigenetic inheritance** involves heritable changes in gene expression — such as DNA methylation patterns — that do not alter the underlying DNA sequence itself. **Mitochondrial DNA** is inherited almost exclusively from the mother, since mitochondria are passed down through the egg cytoplasm.

Why It Matters

Understanding molecular inheritance underpins modern medicine and biotechnology: genetic counseling, prenatal screening, cancer genomics, CRISPR gene editing, and personalized medicine all rest on these foundational principles of how genetic information is stored, expressed, and transmitted across generations.